Upasana Chatterjee

Research Objectives

I am interested in developing principled evaluation methodologies for language models that go beyond outcome-based metrics to assess reasoning processes and internal representations. By systematically identifying risks and failure modes, I hope to advance our understanding of model behavior and enable more interpretable, aligned, and safely deployable AI systems.

Education

Columbia University, New York, NY **Jan. 2025 – Present**
M.S., Computer Science (Thesis Track)[1]

- Thesis: "Metalinguistic Features in Political Ideology Identification"
- Committee: Daniel Bauer (advisor), Smaranda Muresan, Julia Hirschberg, and Elias Bareinboim.

University of California, Berkeley, CA **Aug. 2015 – Aug. 2019**
B.A., Mathematics (Honors); B.A., Computer Science

- Honors Thesis, Department of Mathematics: "Quadratic Forms over p-adic Fields."
- Advisor: Richard Borcherds.

Publications

1. Chatterjee, U. and Bauer, D.(2026). What You Say vs. How You Say It: Metalinguistic Pre-Training for Political News Classification. *In Submission.*
2. Chatterjee, U. (2026). Does Topic Sentiment Cause Perceived Ideology? Comparing Human and LLM Annotations in Political News Articles. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pp. 725–740. https://arxiv.org/abs/2606.06715

Research Experience

Columbia University, New York, NY
Causal AI Lab May 2026 – Present

- With PhD student Arvind Raghavan, under Prof. Elias Bareinboim, study factuality in LLMs from a causal perspective.
- Apply causal discovery to link linguistic features to entailment performance, characterizing LLMs' factual reasoning errors.

Mobile Social Lab Jan. 2026 – Present

- With PhD student Filipp Shelobolin, under Prof. Augustin Chaintreau, research representation learning for auctions.
- Investigate embedding-based auction representations for predicting bidder behavior and outcomes.

[1] Completed part-time while employed full-time as a Senior Software Engineer at Bloomberg L.P.

Thesis, Department of Computer Science Aug. 2025 – May 2026

- "Metalinguistic Features in Political Ideology Identification," advised by Dr. Daniel Bauer.
- Developed a pre-training objective that leads LLMs to learn metalinguistic features (e.g., theme, tone) predictive of political ideology in news.
- Used causal mediation analysis to show LLMs rely more on theme and tone than human annotators, and that fine-tuning does not reliably close the gap.

University of California, Berkeley, CA

Honors Thesis, Department of Mathematics Jan. 2019 – Aug. 2019

- Advised by Prof. Richard Borcherds, studied quadratic forms over p-adic fields.
- Presented an alternate proof of the Hasse–Minkowski theorem.

Directed Reading Program, Department of Mathematics Aug. 2017 – Dec. 2017

- With a graduate-student mentor, researched pairing-based attacks in elliptic-curve cryptography, presenting findings to an undergraduate audience.

Student Mentoring and Research Teams, Department of History Jun. 2017 – Aug. 2017

- "Making Up Our Minds: Engineers and Philosophers Debate Artificial Intelligence, 1945–1989," on how early AI researchers were shaped by academic debates over AI's plausibility.
- Presented findings to program participants, academics, and donors.

Industry Experience

Bloomberg L.P., New York, NY **Jul. 2020 – Present**

Senior Software Engineer

- With Bloomberg's NLP teams, designed and built ASKB, a natural-language interface for Terminal users, and an NLP-backed platform for personalized cross-application search.
- Own work across the full product lifecycle: requirements, system design, production code, and issue triage.
- Conduct 50+ candidate interviews per year, serve on the LGBTQ+ ERG, Automated Testing Guild, and Women in Equities and Market Intelligence committees.

Aon Cyber Solutions, Boston, MA **Sep. 2019 – Jul. 2020**

Cyber Associate

- Conducted digital-forensic acquisitions and examinations of laptops, desktops, servers, and mobile phones for civil litigation, criminal matters, and internal investigations.
- Developed proprietary software to support investigations and streamline workflow.
- Penetration-tested and code-reviewed 20+ web applications for large enterprises (100,000+ employees).

Teaching & Mentoring

PENCIL, Inc., New York, NY **May 2024 – Aug. 2024**

Career Explorers Tech Mentor

- Mentored a first-generation incoming university student through a full-stack project building a browser game to improve financial literacy.
- Provided technical direction and guidance on study skills and engineering practices.

Code Nation, New York, NY **Aug. 2023 – Jun. 2024**
Computer Science Mentor

- Taught software-engineering fundamentals to high-school students in a year-long after-school program.
- Led and assisted three-hour biweekly JavaScript classes for a group of 20 students.

PENCIL, Inc., New York, NY **Aug. 2021 – Dec. 2021**
Computer Science Mentor

- Mentored two high-school students in an introductory programming course, teaching three-hour weekly Python classes and offering academic and career counseling.

FemTech Berkeley, Berkeley, CA **Jan. 2018 – Dec. 2018**
Computer Science Instructor

- Taught two-hour weekly classes, wrote problem sets, and ran pre-exam review sessions.
- Served a campus organization providing small-group tutoring to women in STEM courses.
- Courses: CS61B Data Structures (Spring 2018); CS70 Discrete Mathematics (Fall 2018).

Honors & Awards

ACL Student Research Workshop Travel Grant	2026
Departmental Honors (Mathematics), University of California, Berkeley	2019

Technical Skills

Programming Languages & Tools: Python, C++, JavaScript, Docker, Argo, Git, SQL, Bash/Shell
Frameworks & Libraries: PyTorch, TensorFlow, EconML, pandas/polars, NLTK, spaCy, scikit-learn, Hugging Face Transformers

Languages

English (native), Swedish (native), Bengali (fluent), Spanish (conversational), Hindi (conversational)